

Samuel Cancia

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EDUCATION

University of South Carolina
B.S. Mechanical Engineering

GPA: 3.5
Aug 2024 – May 2028

RESEARCH EXPERIENCE

Integrated Multiphysics and Systems Engineering Laboratory (iMSEL)

Columbia, SC

Undergraduate Research Assistant, Dr. Yi Wang

Mar 2025 – Present

- Co-authored paper submitted to AIAA: “Surface-Object Detection and Tracking from Commercial X-Band Marine Radar Using Adaptive ST-DBSCAN”
- Led multi-site data collection campaigns across South Carolina lakes (Murray, Monticello, Greenwood) totaling 30+ hours of X-band radar data, plus 8 hours of multi-modal sensor data in Charleston Harbor
- Developed adaptive ST-DBSCAN object tracking algorithm with automatic neighborhood radius tuning via k-nearest-neighbor distance statistics for robust vessel detection under nonstationary sea clutter
- Implemented real-time tracking pipeline in Rust achieving 2 sweeps/s on NVIDIA Jetson Orin edge platform, exceeding requirements for 48 RPM radar operation
- Conducted ablation study comparing CFAR detector variants and benchmarked detection-to-track association performance against baseline DBSCAN and standard ST-DBSCAN

Adaptive Real-Time Systems Laboratory (ARTS-Lab)

Columbia, SC

Undergraduate Research Assistant, Dr. Austin Downey

May – Aug 2023

- Investigated piezoelectric-based structural modal damping techniques for vibration suppression on instrumented test beams with applications to dynamic aerospace structures
- Designed and built custom pulse generator and power amplifier circuits to drive piezoelectric substrates for active vibration control experiments
- Performed frequency-domain analysis of accelerometer data to identify structural modal locations and quantify damping effectiveness before and after piezoelectric control activation
- Processed experimental vibration data to validate damping performance metrics and support ongoing research in adaptive structural control systems

PROFESSIONAL EXPERIENCE

South Carolina Army National Guard

Columbia, SC

09R SMP Cadet

Jan 2025 – Present

- Trains as future Army Officer through simultaneous ROTC and National Guard service, executing troop leading procedures and tactical operations while maintaining full-time engineering course load

Koffee Koala

Columbia, SC

Barista

Apr 2023 – Feb 2024

- Managed shift operations including opening and closing procedures, staff supervision, and order expediting in high-volume coffee service environment

LEADERSHIP EXPERIENCE

The Daily Gamecock

Columbia, SC

Assistant Editor, Opinion Section

Aug 2024 – Present

SKILLS

Programming:

Engineering Software:

Python, C++, Rust, MATLAB, Arduino
SolidWorks, Creo, Microsoft Office